

Bacterial Staining

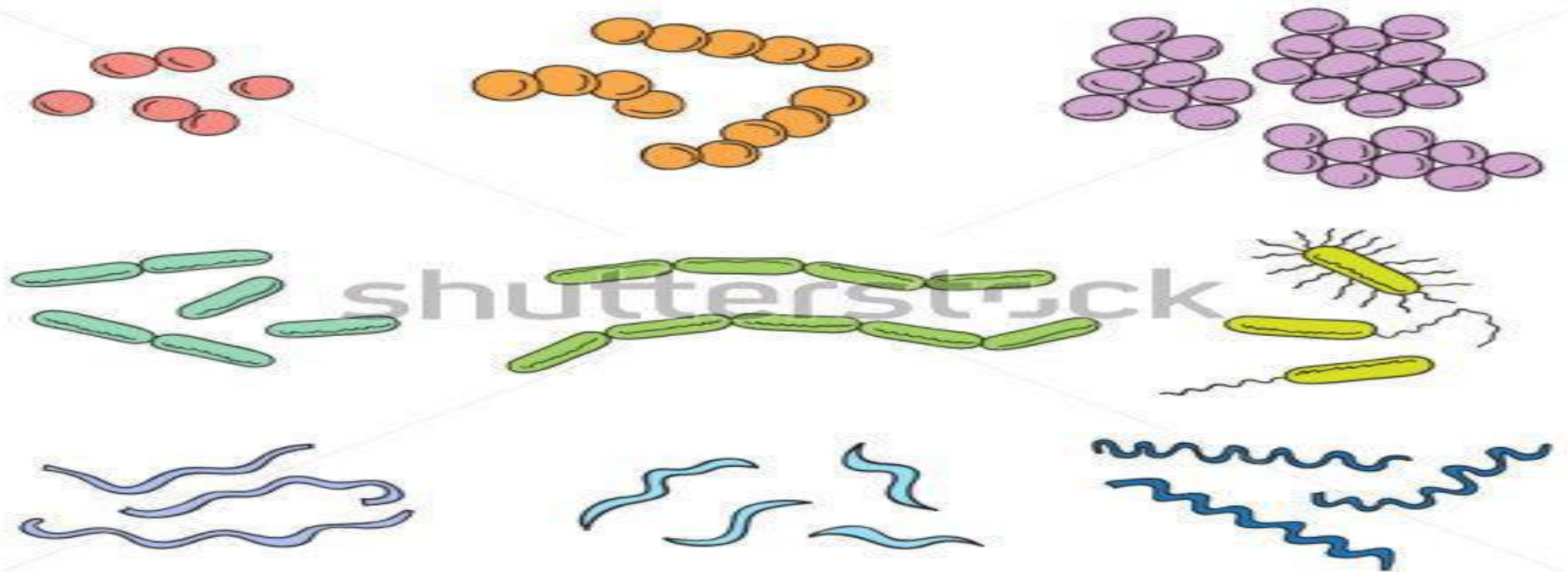
Lab-2

- 1-Simple staining
- 2-Differential staining

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Staining

- There are several staining methods that are used routinely with bacteria. These methods may be classified as:

Simple stain: use only one dye, it will react with different types of bacteria in an identical fashion

Differential stain: use 2 or more dyes, that react differentially with different types bacteria giving varying results depending on the organism being treated

Structural stain: Special stain to examine an internal or external structure of bacterial cell: capsular stain, flagellar stain



Staining

Q: Why do we stain microorganisms?

- Bacteria are microscopic and colorless organisms.
- I. In order to visualize them and study their structure, shape and other structural characteristics, it becomes necessary to make them more easily visible.

Purpose of staining

2-Staining organisms enable us to observe different microbial characteristics.

- –**Shape and arrangement**
- –**Gram reactions**
- –**Endospore production**
- –**Capsule Production**
- –**Pathogenic bacteria**

Types of dyes

I -Basic dye: A positively charged(cationic dye):

- Works best in neutral or alkaline pH.
- Cell wall has slight negative charge at pH 7.
- eg. Crystal violet, methylene blue, safranine

Types of dyes

- **2-Acidic dye:** A negatively charged (an anionic dye)
- Chromophore repelled by -ve cell wall.
- Background is stained, bacteria are colorless.
- •e.g. **Eosin**, **India ink**

Q/ What is the first step in staining?

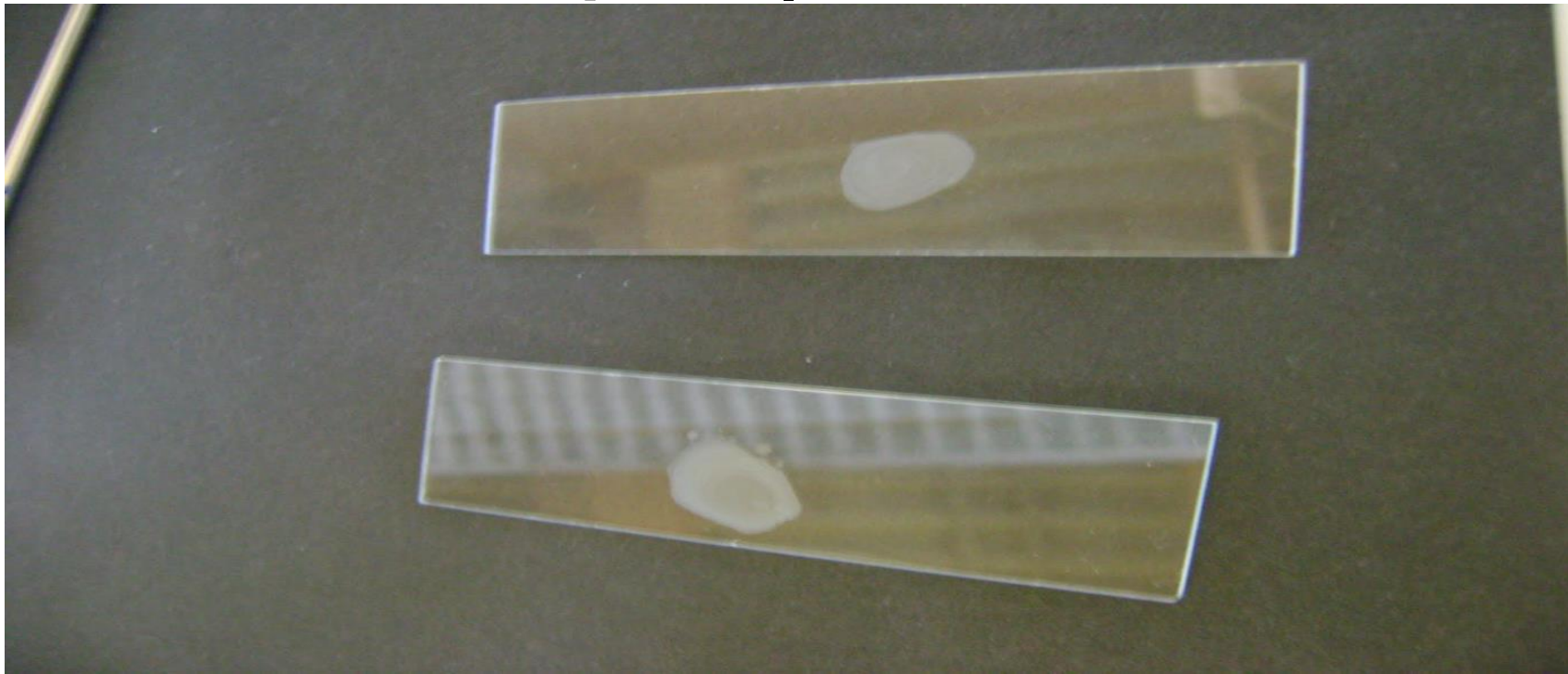
Smear Preparation -----is always the **first step** in any microbial stain.

- It is spreading microbial cells on a clean glass slide and allowing it to air dry and **heat fix**.



Benefit of Heat Fixation

1. Kills the organism.
2. It causes the organisms to adhere to the slide.
3. It alters the organisms so that they more readily accept stains .



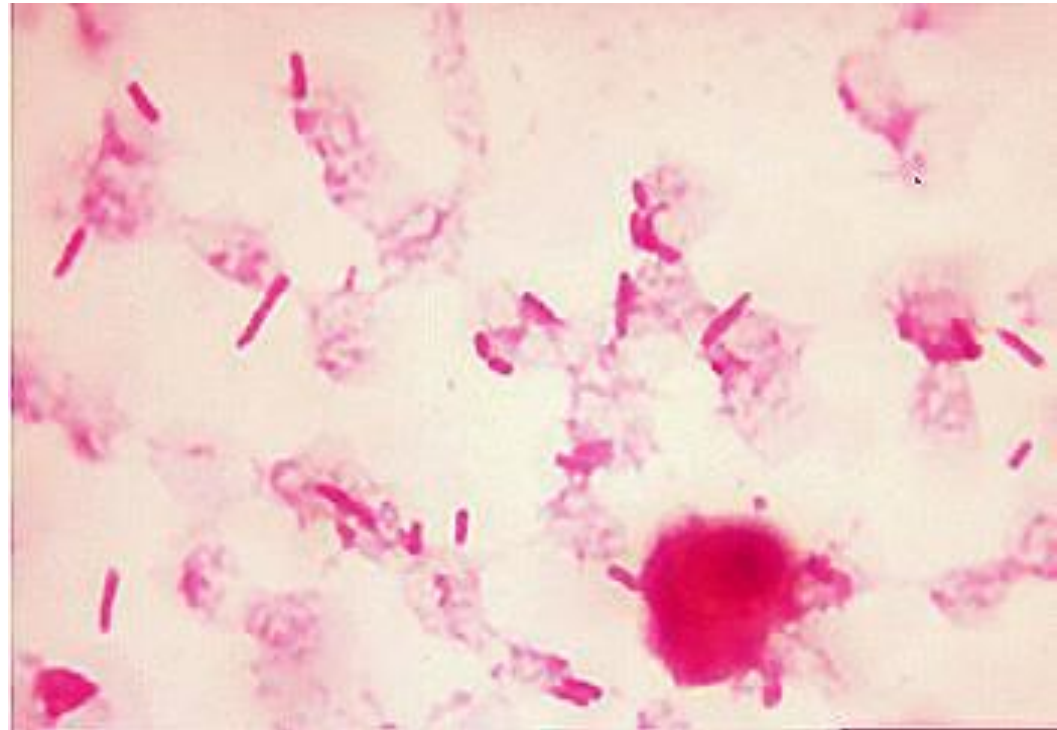
Simple Stain

One dye, one step.

- Direct stain using basic or acidic dye .
- To study the morphology and structure of organisms.
- There are two methods:

1-Positive staining

2-Negative staining



GramStain

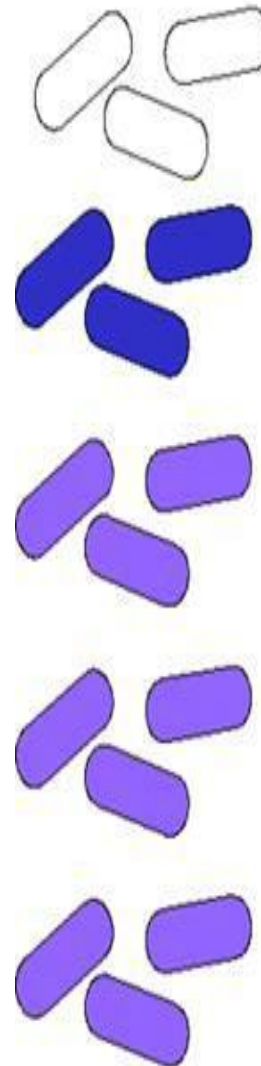
Purpose:

- it is the first step in bacterial identification
- it distinguishes between Gram +ve and Gram –ve cells,
- assists to identify morphology and cellular arrangement.

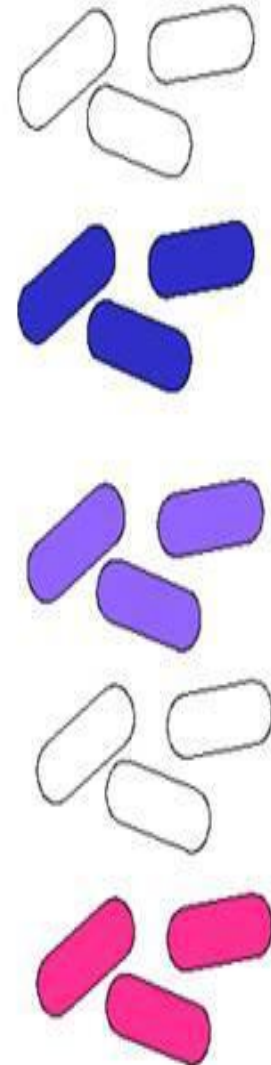
Principle of staining technique:

1. **Crystal violet:** all cells are stained violet.
2. Iodine acts as a **mordant** (fixes the dye) all cells remain violet.
3. Alcohol acetone decolorizes gram negative cells only: Gram-positive remains violet, gram-negative becomes colorless.
4. Counter stain with **Safranin:** Gram-positive remains violet while gram negative becomes red.

Gram Positive



Gram Negative



Fixation

Crystal violet

Iodine treatment

Decolorization

Counter stain
safranin

Gram staining

Materials

1. 24-hour bacterial growth (agar or broth).
2. Crystal violet.....Primary stain
3. Gram's iodine.....Mordant
4. Ethyl alcohol, 95%.....Decolorizing agent
5. Safranin.....Counter stain
6. Slides
7. Marking pen or pencil and slide labels.

Decolorize with alcohol (95%) Be careful not to over decolorize, as many Gram-positive organisms may lose the violet stain easily and thus appear to be Gram negative after they are counterstained.

Procedure

1. **Prepare a fixed smear.**
2. **Flood slide with crystal violet. Allow to stand for one minute .**
3. **Wash off with tap water.**
4. **Flood with Gram's iodine (a mordant). Leave for one minute.**
5. **Wash off with tap water.**
6. **Decolorize with alcohol (95%)(usually 10–20 seconds).**
7. **Wash off with tap water.**
8. **Apply safranin for one minute.**
9. **Wash off with tap water.**
10. **Drain and blot gently with bibulous paper.**
11. **Air dry the slide and examine the preparation under the microscope.**

Staining Techniques

- **Simple Stain**

- Uses only one basic dye
- Provides basic information about cell shape and arrangement

- **Differential Stain**

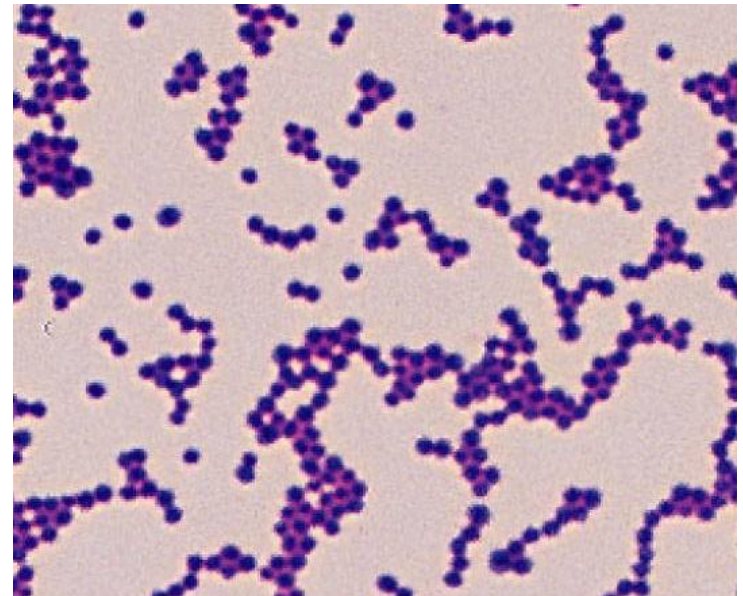
- Uses more than one dye
- These procedures react differently with different kinds of bacteria
- Helps distinguish between different kinds of bacteria
- Most common and important differential stain is the GRAM STAIN

Shape and arrangement

(Streptococci)

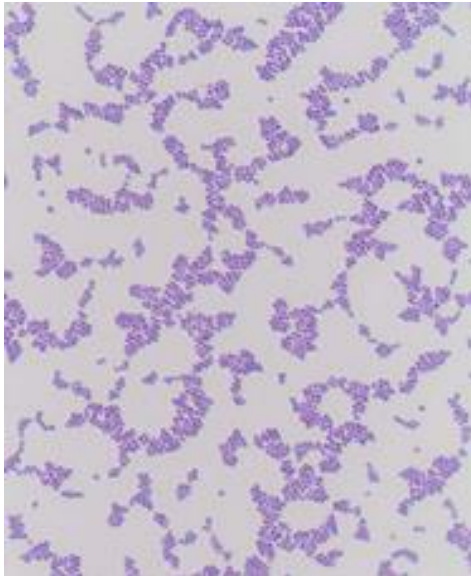


(Staphylococci)

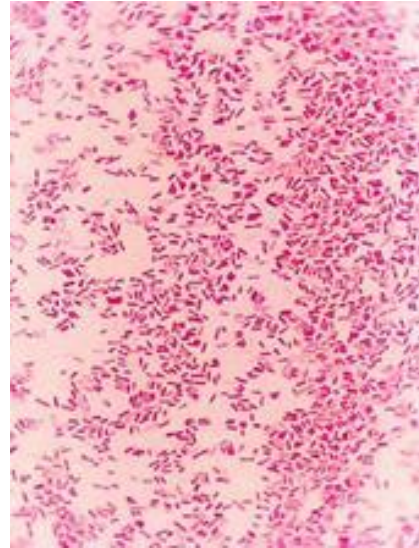


Gram stain result

Staphylococcus



Escherichia coli

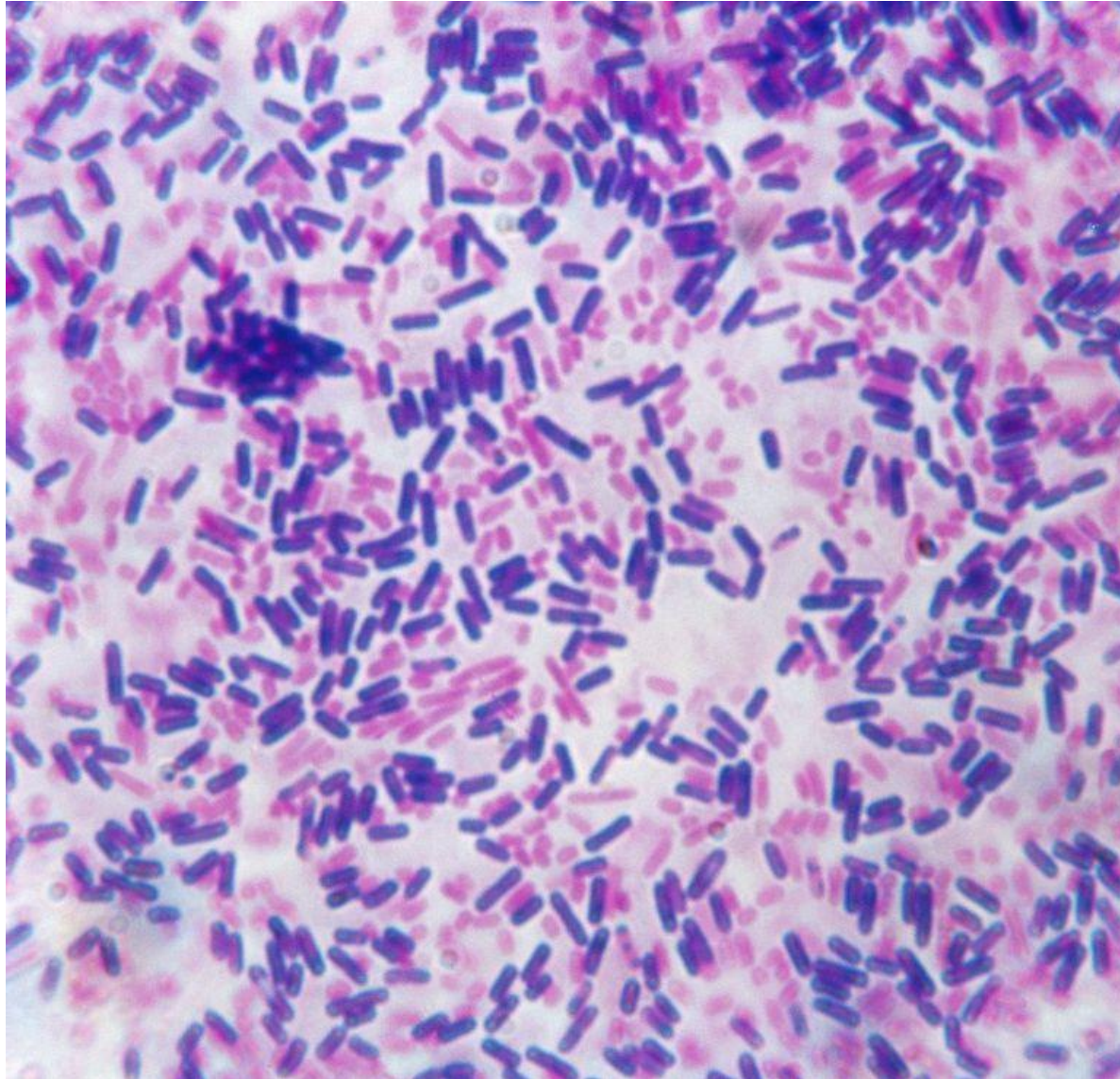


Bacillus subtilis



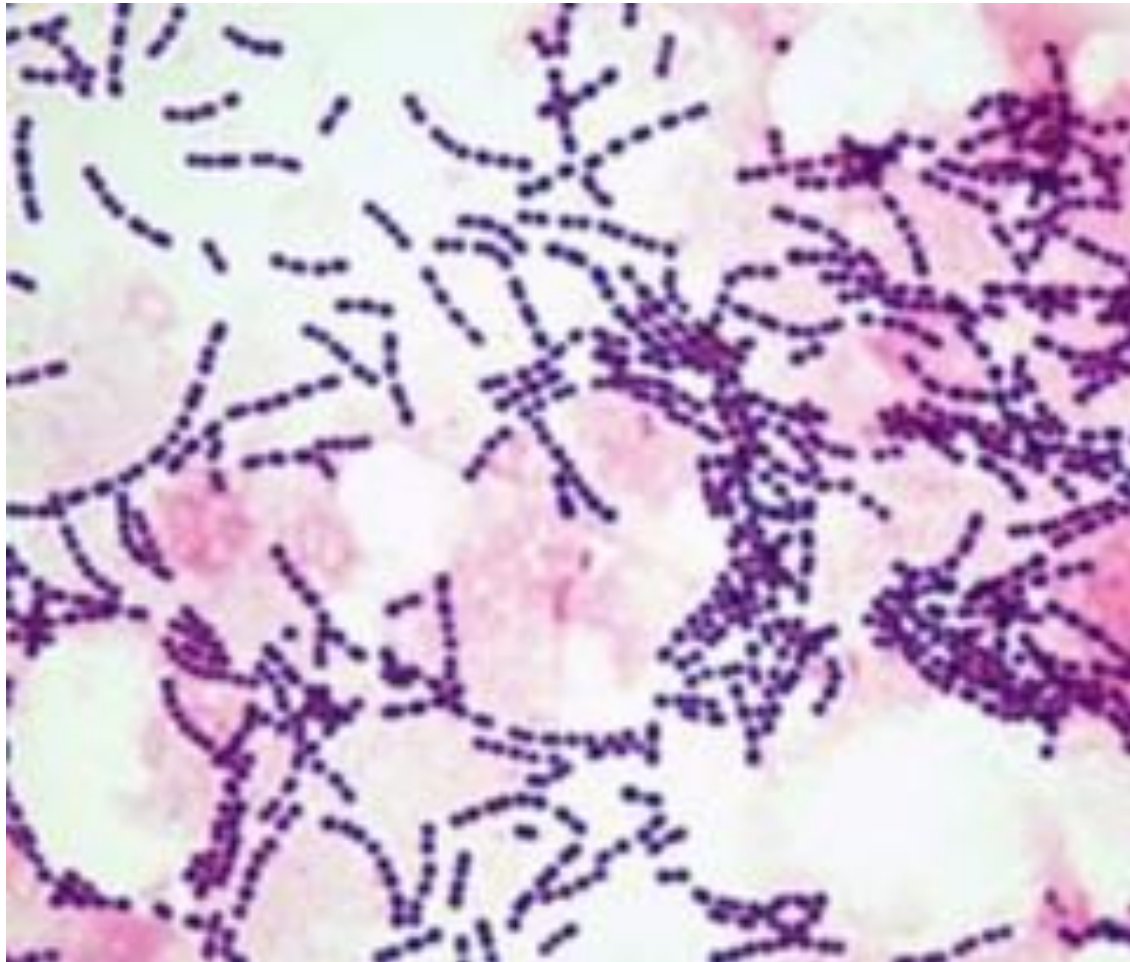
Gram reaction:	Gram-positive	Gram-negative	Gram-positive
Shape:	Cocci	Rods	Rods
Size:	Small	Small	Large
Arrangement:	Clusters	Single	Chains

Gram +ve and gram -ve rods



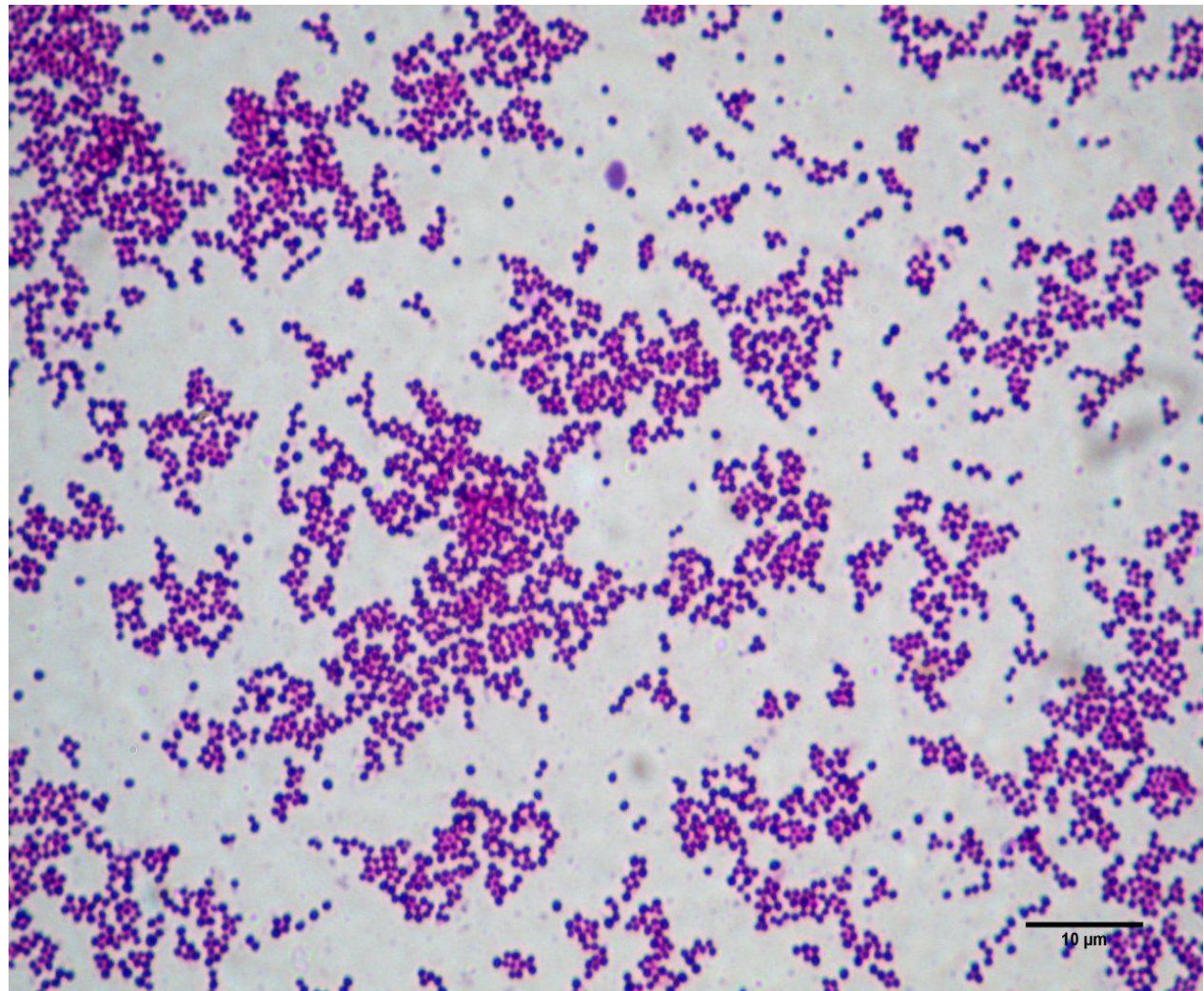
Gram +ve cocci

(*Streptococcus spp.*)

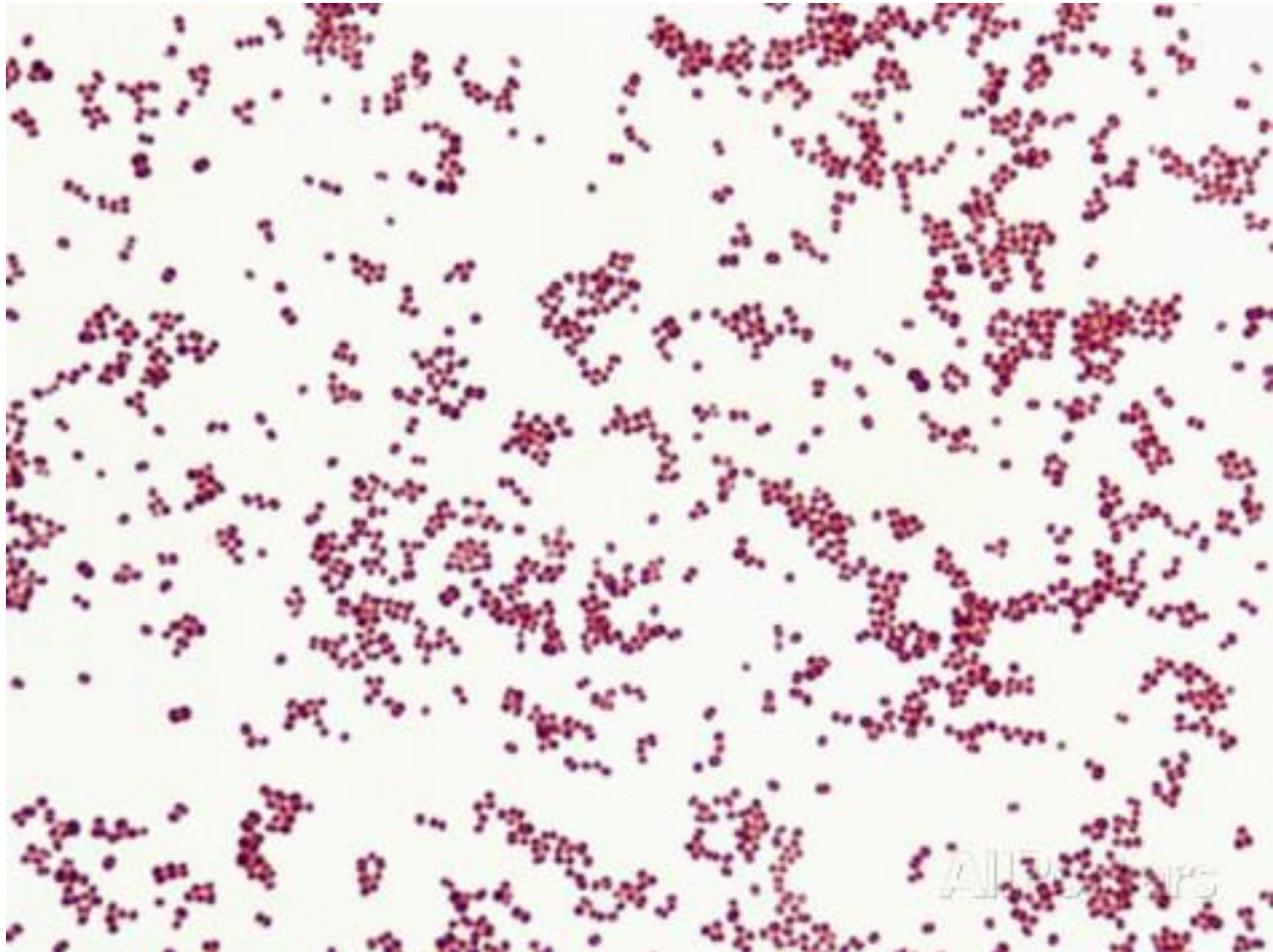


Gram +ve cocci

Staphylococcus spp.



Gram -ve Cocci (*Neisseria* spp.)



**Thank
You**